Azure Operations Dashboard

VM Health & Performance Monitoring — Project Evidence Presentation

VM STATUS	ACTIVE ALERTS	RESOURCE HEALTH
RUNNING	2 ENABLED	AVAILABLE

Resource group: rg-azureops-hackathon | VM: vm-azureops-demo | Region: Central India

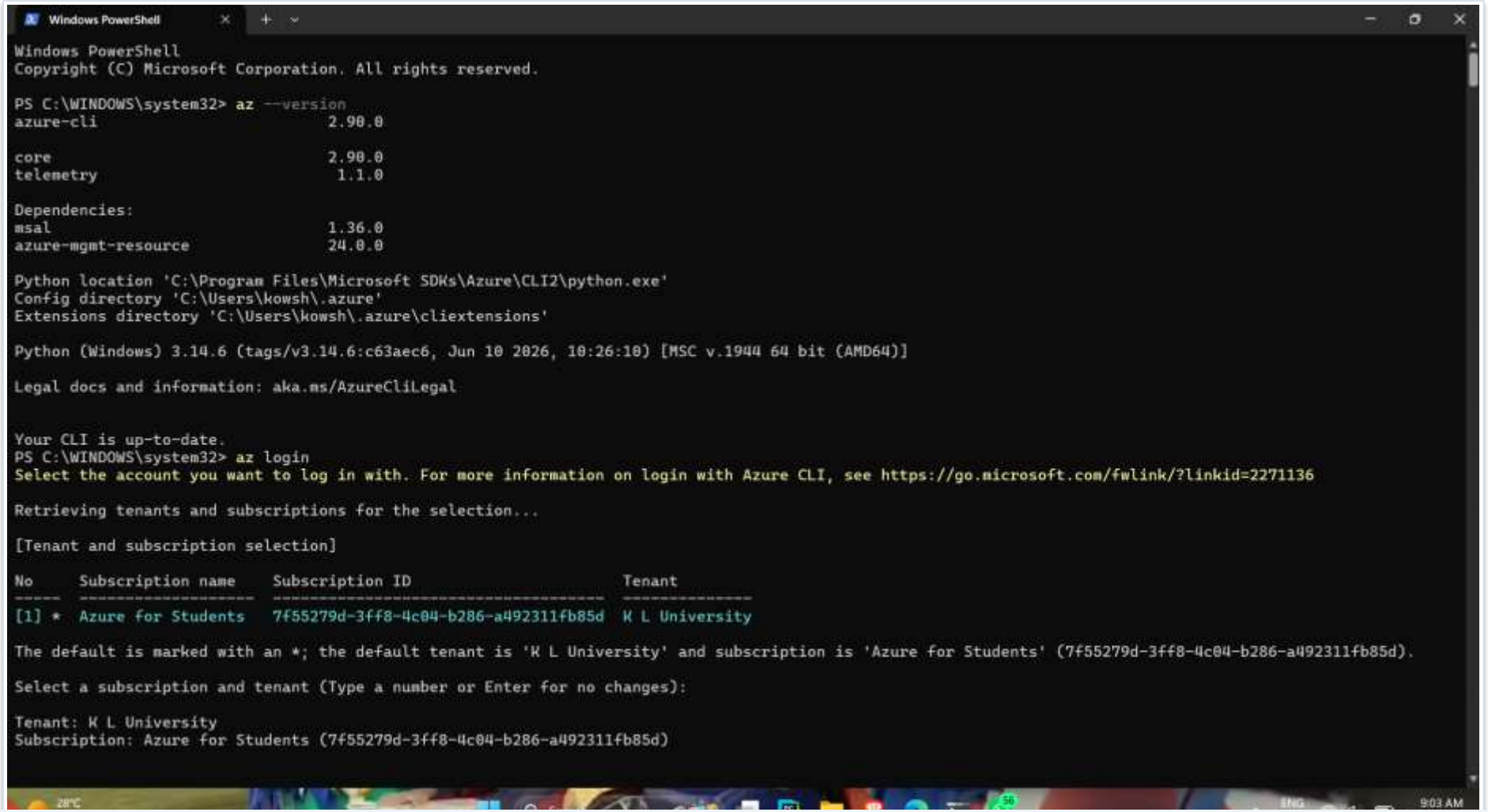
Monitoring: Azure Monitor + OpenTelemetry metrics | Workbook: Azure Operations Dashboard

Evidence covers CLI verification, VM telemetry, alerts, dashboard charts, KPIs and resource health.

Project Overview

Infrastructure	Ubuntu Server 24.04 LTS VM • Standard_B2ats_v2 • Central India
Monitoring	Azure Monitor metrics • OpenTelemetry • Log Analytics workspace
Alerts	Percentage CPU > 80 • VM Availability < 1 • Email notification configured
Dashboard	Operational Workbook with charts, active alerts, KPIs and Resource Health
Verification	Azure CLI resource inventory, instance state and metric queries

Azure CLI Login & Subscription



Azure CLI 2.90.0 is installed and the Azure for Students subscription is selected.

Azure Resources & VM Status

```
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://g
o.microsoft.com/fwlink/?linkid=2271236

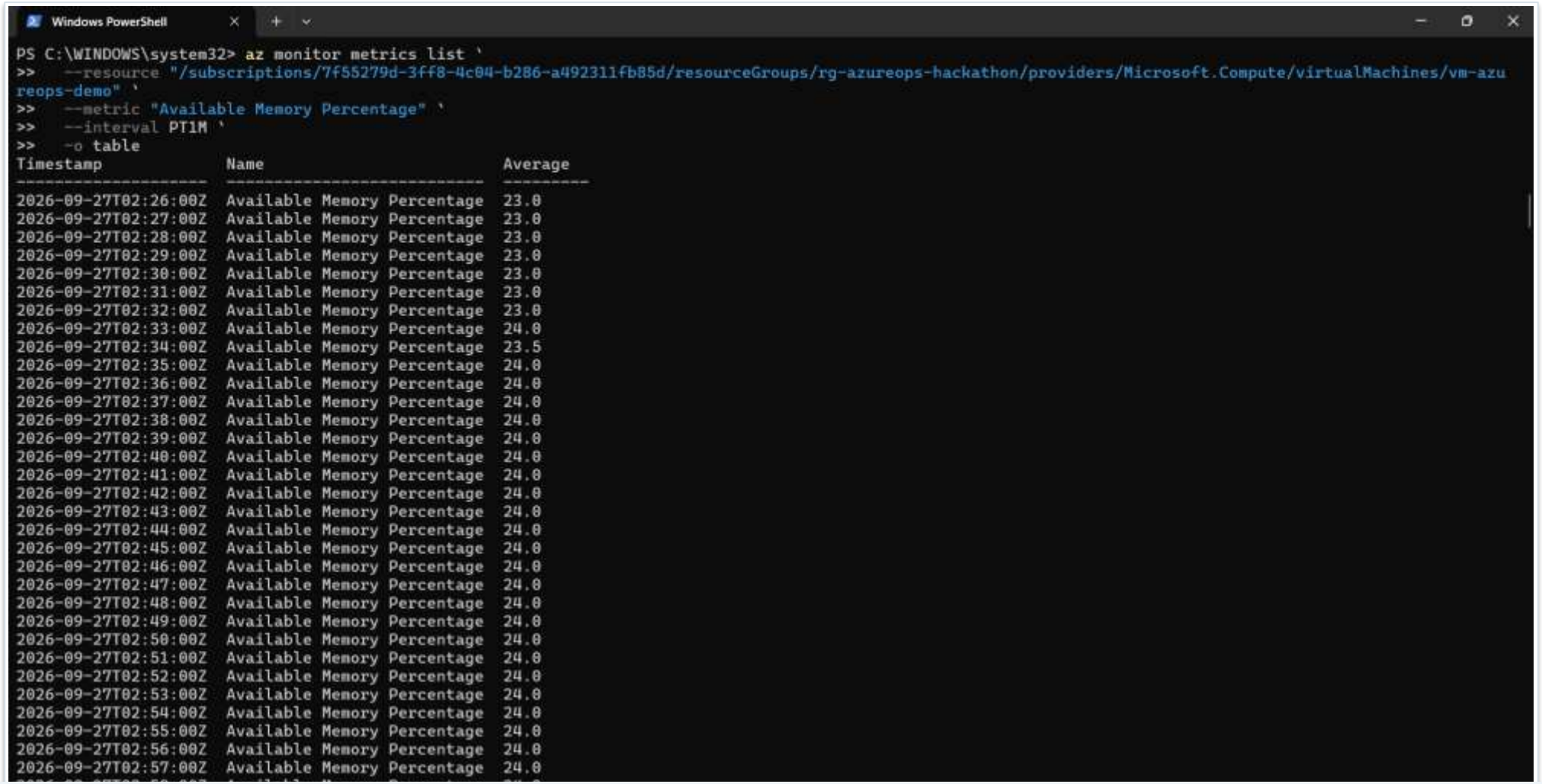
If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

PS C:\WINDOWS\system32> az resource list -g rg-azureops-hackathon -o table
Name                                                 ResourceGroup          Location      Type                                                Status
---------------------------------------------------  ---------------------  ------------  --------------------------------------------------  ---------
law-azureops-hackathon                               rg-azureops-hackathon  centralindia  Microsoft.OperationalInsights/workspaces            Succeeded
vm-azureops-demo_key                                 rg-azureops-hackathon  centralindia  Microsoft.Compute/sshPublicKeys                     Succeeded
vm-azureops-demo-nsg                                 rg-azureops-hackathon  centralindia  Microsoft.Network/networkSecurityGroups             Succeeded
vm-azureops-demo-ip                                  rg-azureops-hackathon  centralindia  Microsoft.Network/publicIPAddresses                 Succeeded
vnet-centralindia-1                                  rg-azureops-hackathon  centralindia  Microsoft.Network/virtualNetworks                   Succeeded
vm-azureops-demo62                                   rg-azureops-hackathon  centralindia  Microsoft.Network/networkInterfaces                 Succeeded
vm-azureops-demo                                     rg-azureops-hackathon  centralindia  Microsoft.Compute/virtualMachines                   Succeeded
vm-azureops-demo_disk1_125d3af953714059b79911f6929b2fff  RG-AZUREOPS-HACKATHON  centralindia  Microsoft.Compute/disks                             Succeeded
msvmi-centralindia-vm-azureops-demo                  rg-azureops-hackathon  centralindia  Microsoft.Insights/dataCollectionRules              Succeeded
vm-azureops-demo/AzureMonitorLinuxAgent              rg-azureops-hackathon  centralindia  microsoft.compute/virtualMachines/extensions        Succeeded
21d4cd0c-2fd6-49e6-9654-ee084de76010                 rg-azureops-hackathon  centralindia  microsoft.insights/workbooks                        Succeeded
RecommendedAlertRules-AG-1                           rg-azureops-hackathon  global        microsoft.insights/actiongroups                     Succeeded
VM Availability - vm-azureops-demo                   rg-azureops-hackathon  global        Microsoft.Insights/metricalerts                     Succeeded
Percentage CPU - vm-azureops-demo                    rg-azureops-hackathon  global        Microsoft.Insights/metricalerts                     Succeeded
PS C:\WINDOWS\system32> az vm get-instance-view -g rg-azureops-hackathon -n vm-azureops-demo --query "instanceView.statuses[1]" -o table
Code                DisplayStatus    Level
------------------  ---------------  -------
PowerState/running  VM running       Info
PS C:\WINDOWS\system32> az monitor metrics list `
>>   --resource "/subscriptions/7f55279d-3ff8-4c04-b286-a492311fb85d/resourceGroups/rg-azureops-hackathon/providers/Microsoft.Compute/virtualMachines/vm-azu
reops-demo" `
>>   --metric "Percentage CPU" `
>>   --interval PT1M `
>>   -o table
Timestamp            Name            Average
-------------------  --------------  ---------
2026-09-27T02:25:00Z  Percentage CPU  0.6
2026-09-27T02:26:00Z  Percentage CPU  0.585
2026-09-27T02:27:00Z  Percentage CPU  0.605
2026-09-27T02:28:00Z  Percentage CPU  0.59
```

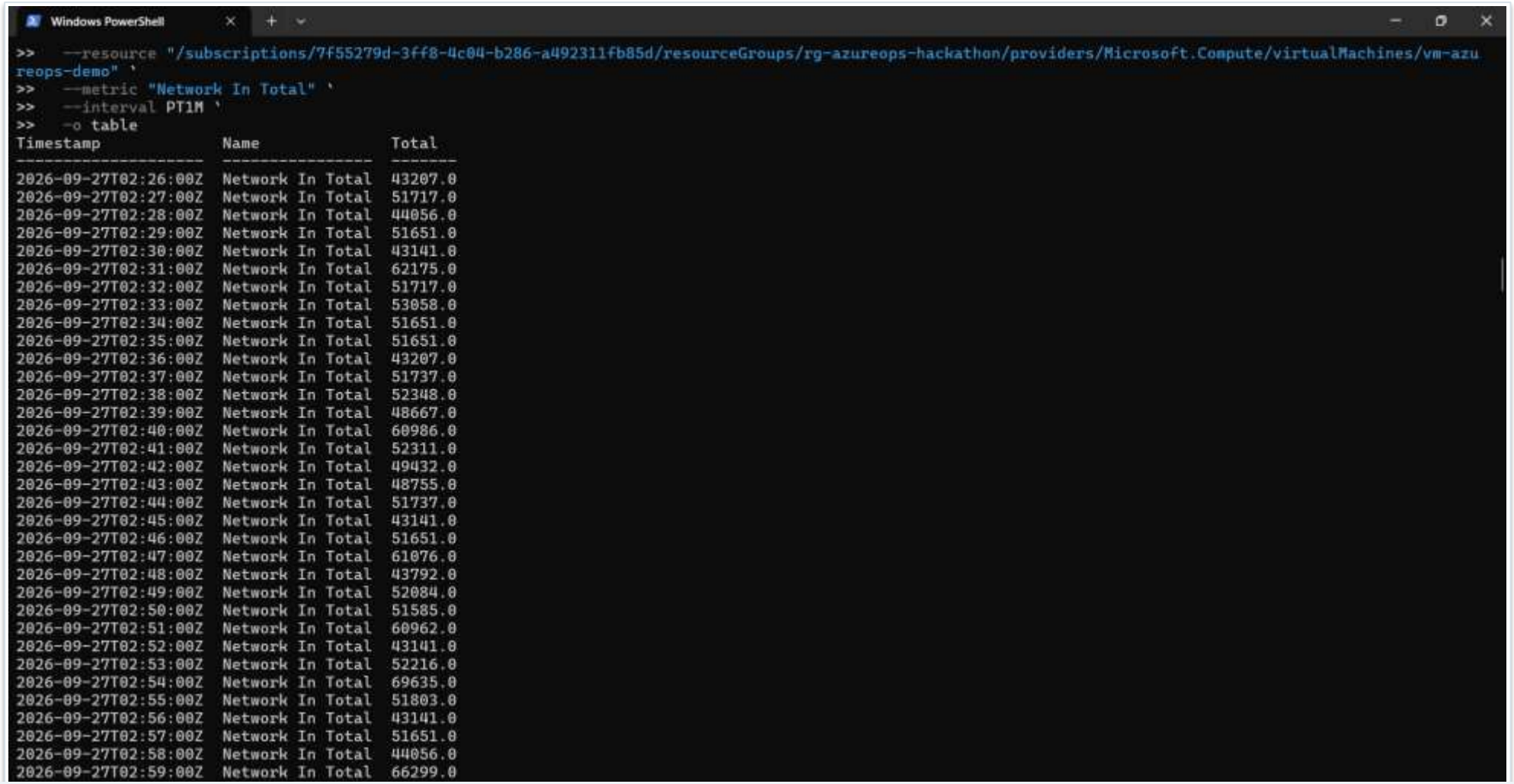
Resource inventory shows the monitoring stack, VM, networking, workbook and metric alerts. The VM is running.

CPU Telemetry



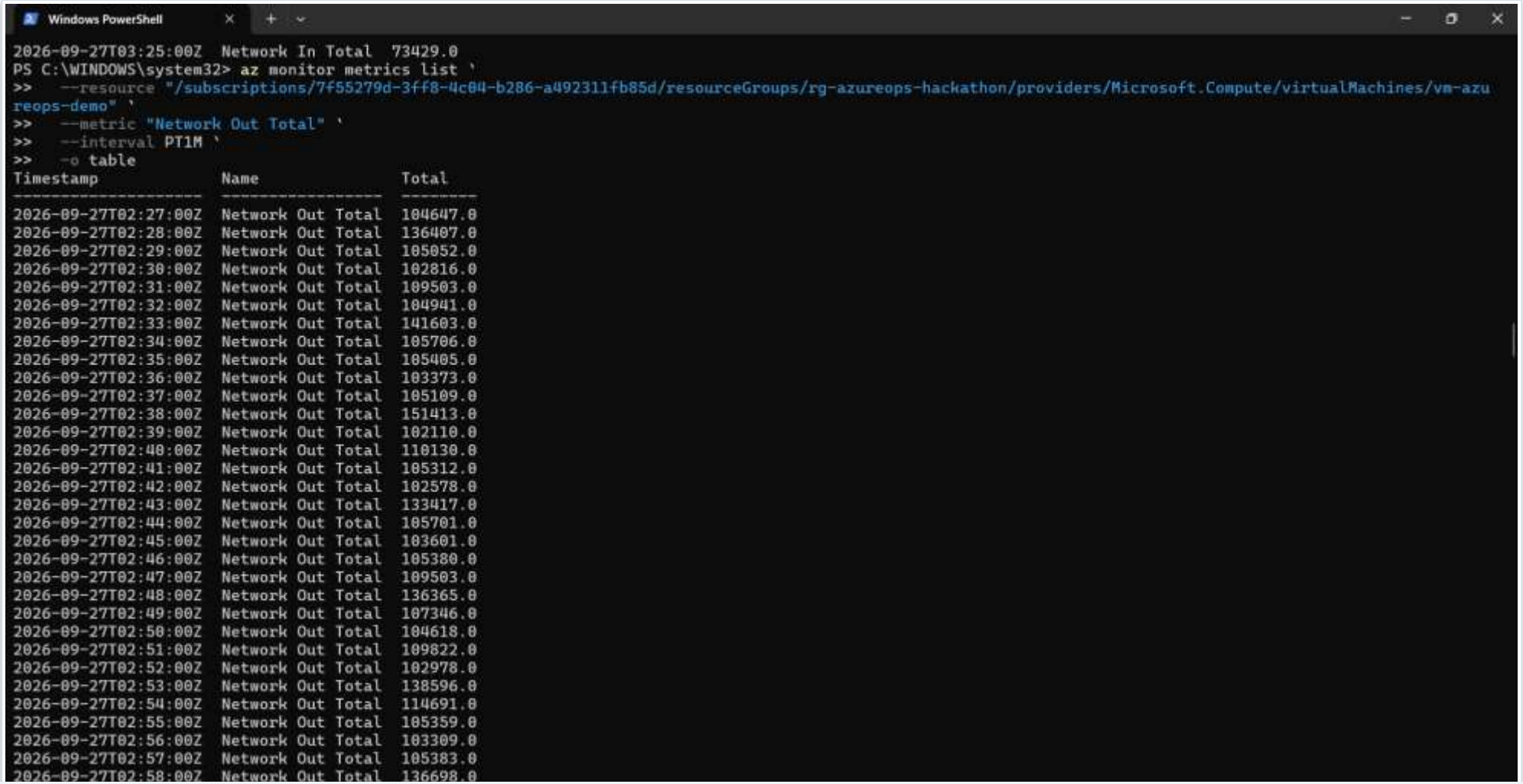
Minute-level Percentage CPU telemetry is available, with the captured values around 0.6%.

Available Memory Telemetry



Available Memory Percentage is continuously reported, around 23–24% in the captured interval.

Network In Telemetry



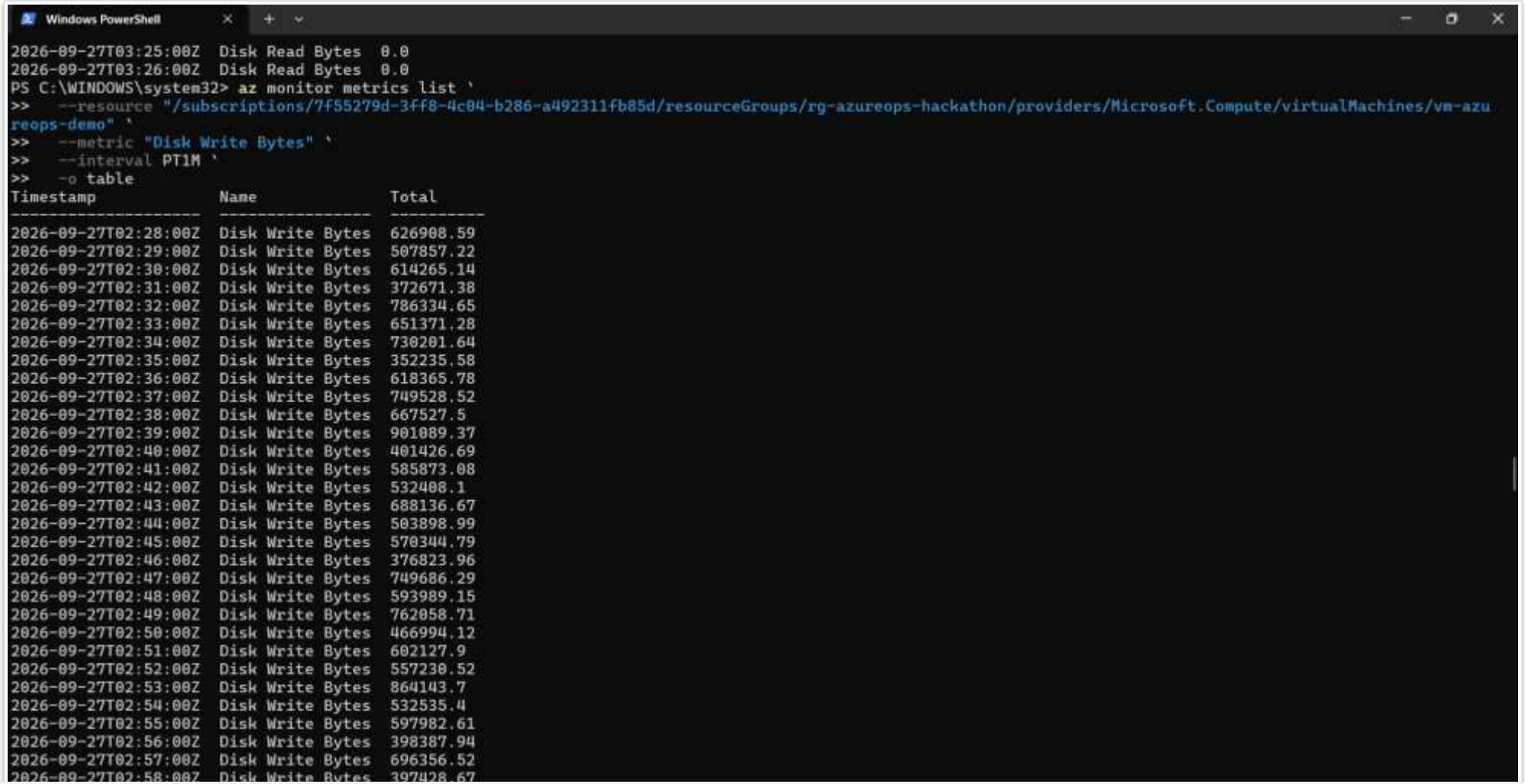
Network In Total provides continuous inbound traffic measurements.

Network Out Telemetry

```
PS C:\WINDOWS\system32> az monitor metrics list `
>>   --resource "/subscriptions/7f55279d-3ff8-4c04-b286-a492311fb85d/resourceGroups/rg-azureops-hackathon/providers/Microsoft.Compute/virtualMachines/vm-azu
reops-demo" `
>>   --metric "Disk Read Bytes" `
>>   --interval PT1M `
>>   -o table
Timestamp            Name             Total
-------------------  ---------------  ----------
2026-09-27T02:27:00Z  Disk Read Bytes  0.0
2026-09-27T02:28:00Z  Disk Read Bytes  0.0
2026-09-27T02:29:00Z  Disk Read Bytes  0.0
2026-09-27T02:30:00Z  Disk Read Bytes  0.0
2026-09-27T02:31:00Z  Disk Read Bytes  0.0
2026-09-27T02:32:00Z  Disk Read Bytes  0.0
2026-09-27T02:33:00Z  Disk Read Bytes  0.0
2026-09-27T02:34:00Z  Disk Read Bytes  0.0
2026-09-27T02:35:00Z  Disk Read Bytes  0.0
2026-09-27T02:36:00Z  Disk Read Bytes  0.0
2026-09-27T02:37:00Z  Disk Read Bytes  0.0
2026-09-27T02:38:00Z  Disk Read Bytes  0.0
2026-09-27T02:39:00Z  Disk Read Bytes  0.0
2026-09-27T02:40:00Z  Disk Read Bytes  0.0
2026-09-27T02:41:00Z  Disk Read Bytes  0.0
2026-09-27T02:42:00Z  Disk Read Bytes  0.0
2026-09-27T02:43:00Z  Disk Read Bytes  0.0
2026-09-27T02:44:00Z  Disk Read Bytes  0.0
2026-09-27T02:45:00Z  Disk Read Bytes  0.0
2026-09-27T02:46:00Z  Disk Read Bytes  0.0
2026-09-27T02:47:00Z  Disk Read Bytes  0.0
2026-09-27T02:48:00Z  Disk Read Bytes  0.0
2026-09-27T02:49:00Z  Disk Read Bytes  0.0
2026-09-27T02:50:00Z  Disk Read Bytes  0.0
2026-09-27T02:51:00Z  Disk Read Bytes  0.0
2026-09-27T02:52:00Z  Disk Read Bytes  0.0
2026-09-27T02:53:00Z  Disk Read Bytes  0.0
2026-09-27T02:54:00Z  Disk Read Bytes  0.0
2026-09-27T02:55:00Z  Disk Read Bytes  0.0
```

Network Out Total provides continuous outbound traffic measurements.

Disk Read Bytes



Disk Read Bytes telemetry is available; the captured interval shows very low read activity.

Disk Write Bytes

```
2026-09-27T03:27:00Z  Disk Write Bytes
PS C:\WINDOWS\system32> az monitor metrics list `
>>   --resource "/subscriptions/7f55279d-3ff8-4c04-b286-a492311fb85d/resourceGroups/rg-azureops-hackathon/providers/Microsoft.Compute/virtualMachines/vm-azu
reops-demo" `
>>   --metric "Disk Read Operations/Sec" `
>>   --interval PT1M `
>>   -o table
Timestamp            Name                      Average
-------------------  ------------------------  ---------
2026-09-27T02:28:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:29:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:30:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:31:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:32:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:33:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:34:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:35:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:36:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:37:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:38:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:39:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:40:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:41:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:42:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:43:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:44:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:45:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:46:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:47:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:48:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:49:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:50:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:51:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:52:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:53:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:54:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:55:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:56:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:57:00Z  Disk Read Operations/Sec  0.0
2026-09-27T02:58:00Z  Disk Read Operations/Sec  0.0
```

Disk Write Bytes shows sustained disk write activity across the captured minutes.

Disk Read Operations / Sec

```
PS C:\WINDOWS\system32> az monitor metrics list `
>>   --resource "/subscriptions/7f55279d-3ff8-4c04-b286-a492311fb85d/resourceGroups/rg-azureops-hackathon/providers/Microsoft.Compute/virtualMachines/vm-azu
reops-demo" `
>>   --metric "Disk Write Operations/Sec" `
>>   --interval PT1M `
>>   -o table
Timestamp            Name                        Average
-------------------  --------------------------  ---------
2026-09-27T02:29:00Z  Disk Write Operations/Sec  0.615
2026-09-27T02:30:00Z  Disk Write Operations/Sec  0.915
2026-09-27T02:31:00Z  Disk Write Operations/Sec  0.565
2026-09-27T02:32:00Z  Disk Write Operations/Sec  1.145
2026-09-27T02:33:00Z  Disk Write Operations/Sec  0.99
2026-09-27T02:34:00Z  Disk Write Operations/Sec  1.0
2026-09-27T02:35:00Z  Disk Write Operations/Sec  0.52
2026-09-27T02:36:00Z  Disk Write Operations/Sec  0.905
2026-09-27T02:37:00Z  Disk Write Operations/Sec  1.13
2026-09-27T02:38:00Z  Disk Write Operations/Sec  0.945
2026-09-27T02:39:00Z  Disk Write Operations/Sec  1.215
2026-09-27T02:40:00Z  Disk Write Operations/Sec  0.605
2026-09-27T02:41:00Z  Disk Write Operations/Sec  0.915
2026-09-27T02:42:00Z  Disk Write Operations/Sec  0.78
2026-09-27T02:43:00Z  Disk Write Operations/Sec  0.98
2026-09-27T02:44:00Z  Disk Write Operations/Sec  0.665
2026-09-27T02:45:00Z  Disk Write Operations/Sec  0.91
2026-09-27T02:46:00Z  Disk Write Operations/Sec  0.555
2026-09-27T02:47:00Z  Disk Write Operations/Sec  1.13
2026-09-27T02:48:00Z  Disk Write Operations/Sec  0.88
2026-09-27T02:49:00Z  Disk Write Operations/Sec  1.0
2026-09-27T02:50:00Z  Disk Write Operations/Sec  0.67
2026-09-27T02:51:00Z  Disk Write Operations/Sec  0.915
2026-09-27T02:52:00Z  Disk Write Operations/Sec  0.8
2026-09-27T02:53:00Z  Disk Write Operations/Sec  1.205
2026-09-27T02:54:00Z  Disk Write Operations/Sec  0.67
2026-09-27T02:55:00Z  Disk Write Operations/Sec  0.945
2026-09-27T02:56:00Z  Disk Write Operations/Sec  0.625
2026-09-27T02:57:00Z  Disk Write Operations/Sec  1.08
2026-09-27T02:58:00Z  Disk Write Operations/Sec  0.565
2026-09-27T02:59:00Z  Disk Write Operations/Sec  0.945
2026-09-27T03:00:00Z  Disk Write Operations/Sec  0.675
2026-09-27T03:01:00Z  Disk Write Operations/Sec  0.94
```

Disk Read Operations/Sec is available and is almost entirely 0.0 in the displayed interval.

Disk Write Operations / Sec

```
2026-09-27T03:28:00Z  Disk Write Operations/Sec
PS C:\WINDOWS\system32> az monitor metrics list `
>>   --resource "/subscriptions/7f55279d-3ff8-4c04-b286-a492311fb85d/resourceGroups/rg-azureops-hackathon/providers/Microsoft.Compute/virtualMachines/vm-azu
reops-demo" `
>>   --metric "CPU Credits Remaining" `
>>   --interval PT1M `
>>   -o table
Timestamp            Name                  Average
-------------------  --------------------  ---------
2026-09-27T02:29:00Z  CPU Credits Remaining  345.28
2026-09-27T02:30:00Z  CPU Credits Remaining  345.67
2026-09-27T02:31:00Z  CPU Credits Remaining  346.06
2026-09-27T02:32:00Z  CPU Credits Remaining  346.45
2026-09-27T02:33:00Z  CPU Credits Remaining  346.84
2026-09-27T02:34:00Z  CPU Credits Remaining  347.23
2026-09-27T02:35:00Z  CPU Credits Remaining  347.62
2026-09-27T02:36:00Z  CPU Credits Remaining  348.01
2026-09-27T02:37:00Z  CPU Credits Remaining  348.4
2026-09-27T02:38:00Z  CPU Credits Remaining  348.79
2026-09-27T02:39:00Z  CPU Credits Remaining  349.18
2026-09-27T02:40:00Z  CPU Credits Remaining  349.57
2026-09-27T02:41:00Z  CPU Credits Remaining  349.96
2026-09-27T02:42:00Z  CPU Credits Remaining  350.35
2026-09-27T02:43:00Z  CPU Credits Remaining  350.74
2026-09-27T02:44:00Z  CPU Credits Remaining  351.13
2026-09-27T02:45:00Z  CPU Credits Remaining  351.52
2026-09-27T02:46:00Z  CPU Credits Remaining  351.91
2026-09-27T02:47:00Z  CPU Credits Remaining  352.3
2026-09-27T02:48:00Z  CPU Credits Remaining  352.69
2026-09-27T02:49:00Z  CPU Credits Remaining  353.08
2026-09-27T02:50:00Z  CPU Credits Remaining  353.47
2026-09-27T02:51:00Z  CPU Credits Remaining  353.86
2026-09-27T02:52:00Z  CPU Credits Remaining  354.25
2026-09-27T02:53:00Z  CPU Credits Remaining  354.64
2026-09-27T02:54:00Z  CPU Credits Remaining  355.03
2026-09-27T02:55:00Z  CPU Credits Remaining  355.42
2026-09-27T02:56:00Z  CPU Credits Remaining  355.81
2026-09-27T02:57:00Z  CPU Credits Remaining  356.2
2026-09-27T02:58:00Z  CPU Credits Remaining  356.59
2026-09-27T02:59:00Z  CPU Credits Remaining  356.98
```

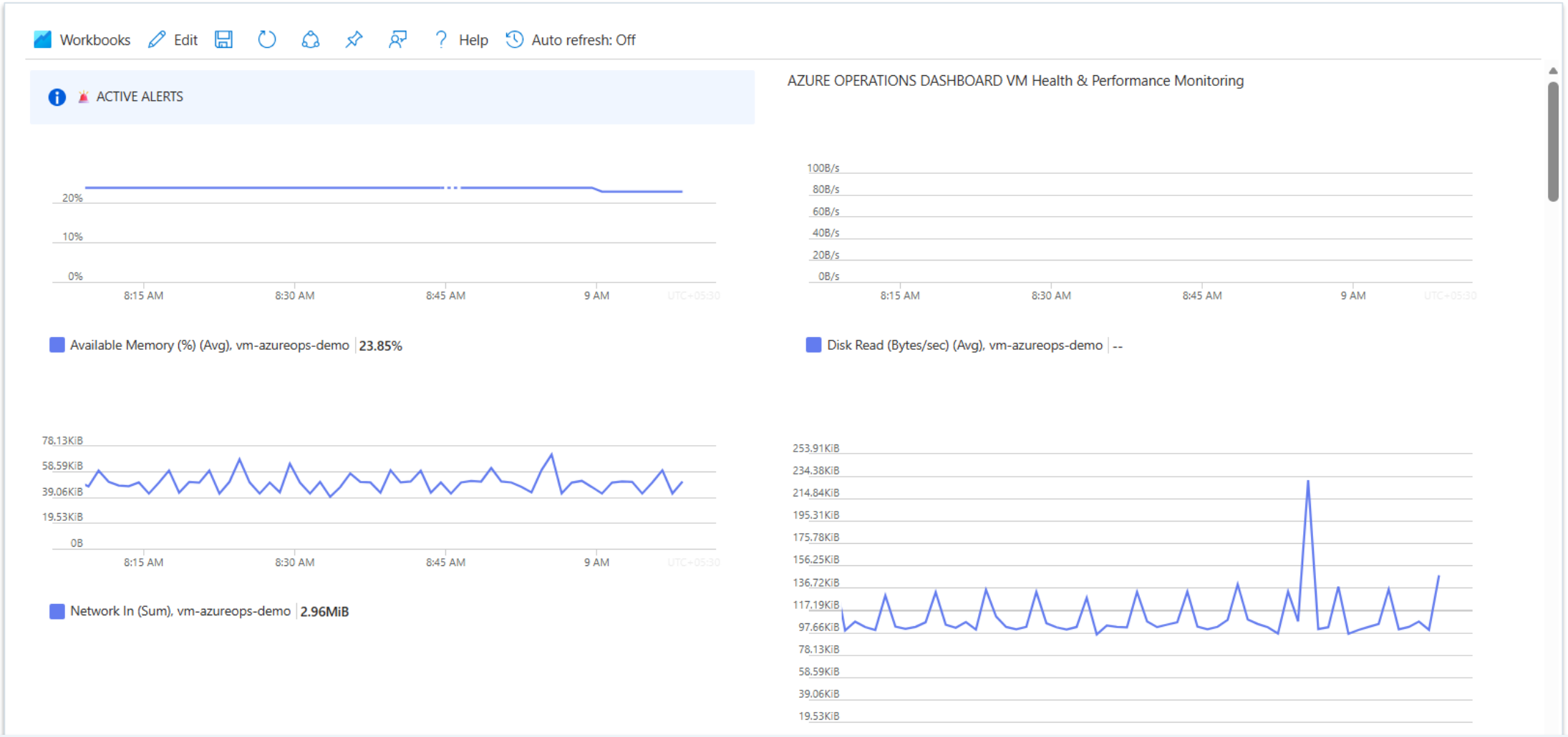
Disk Write Operations/Sec shows regular write activity, generally below about 1.5 ops/sec.

CPU Credits Remaining

```
2026-09-27T03:28:00Z  CPU Credits Remaining  368.28
PS C:\WINDOWS\system32> az monitor metrics list `
>>   --resource "/subscriptions/7f55279d-3ff8-4c04-b286-a492311fb85d/resourceGroups/rg-azureops-hackathon/providers/Microsoft.Compute/virtualMachines/vm-azureops-demo" `
>>   --metric "VmAvailabilityMetric" `
>>   --interval PT1M `
>>   -o table
Timestamp            Name                     Average
-------------------  -----------------------  ---------
2026-09-27T02:31:00Z  VM Availability Metric  1.0
2026-09-27T02:32:00Z  VM Availability Metric  1.0
2026-09-27T02:33:00Z  VM Availability Metric  1.0
2026-09-27T02:34:00Z  VM Availability Metric  1.0
2026-09-27T02:35:00Z  VM Availability Metric  1.0
2026-09-27T02:36:00Z  VM Availability Metric  1.0
2026-09-27T02:37:00Z  VM Availability Metric  1.0
2026-09-27T02:38:00Z  VM Availability Metric  1.0
2026-09-27T02:39:00Z  VM Availability Metric  1.0
2026-09-27T02:40:00Z  VM Availability Metric  1.0
2026-09-27T02:41:00Z  VM Availability Metric  1.0
2026-09-27T02:42:00Z  VM Availability Metric  1.0
2026-09-27T02:43:00Z  VM Availability Metric  1.0
2026-09-27T02:44:00Z  VM Availability Metric  1.0
2026-09-27T02:45:00Z  VM Availability Metric  1.0
2026-09-27T02:46:00Z  VM Availability Metric  1.0
2026-09-27T02:47:00Z  VM Availability Metric  1.0
2026-09-27T02:48:00Z  VM Availability Metric  1.0
2026-09-27T02:49:00Z  VM Availability Metric  1.0
2026-09-27T02:50:00Z  VM Availability Metric  1.0
2026-09-27T02:51:00Z  VM Availability Metric  1.0
2026-09-27T02:52:00Z  VM Availability Metric  1.0
2026-09-27T02:53:00Z  VM Availability Metric  1.0
2026-09-27T02:54:00Z  VM Availability Metric  1.0
2026-09-27T02:55:00Z  VM Availability Metric  1.0
2026-09-27T02:56:00Z  VM Availability Metric  1.0
2026-09-27T02:57:00Z  VM Availability Metric  1.0
2026-09-27T02:58:00Z  VM Availability Metric  1.0
2026-09-27T02:59:00Z  VM Availability Metric  1.0
2026-09-27T03:00:00Z  VM Availability Metric  1.0
2026-09-27T03:01:00Z  VM Availability Metric  1.0
```

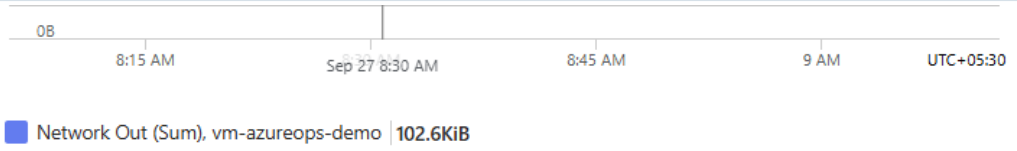
CPU Credits Remaining is continuously reported and increases during the low-CPU interval.

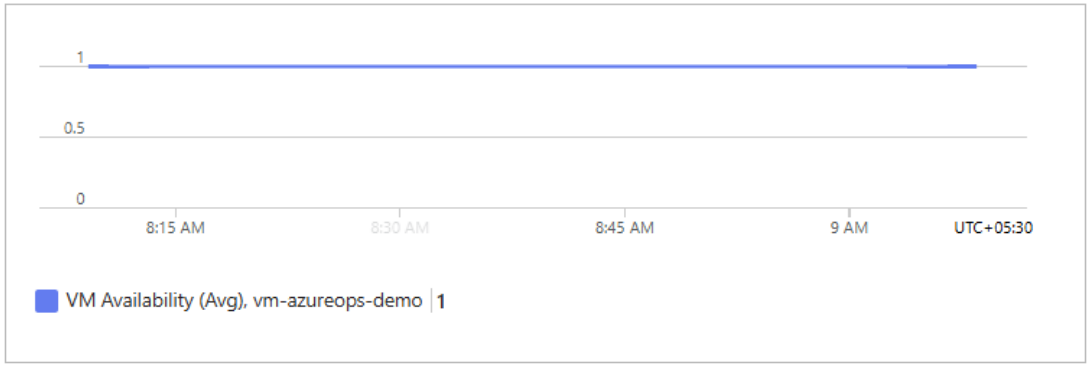
VM Availability Metric



VM Availability Metric remains at 1.0 throughout the displayed period.

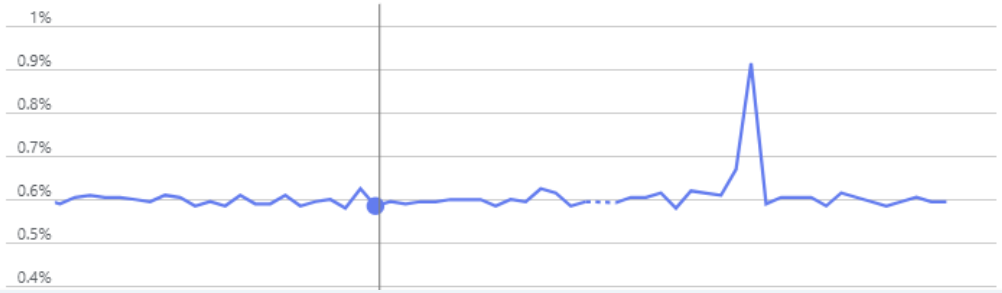
Dashboard — Alerts & Monitoring





AlertName	Enabled	Severity	Description
Percentage CPU - vm-azureops-demo	true	3	
VM Availability - vm-azureops-demo	true	3	

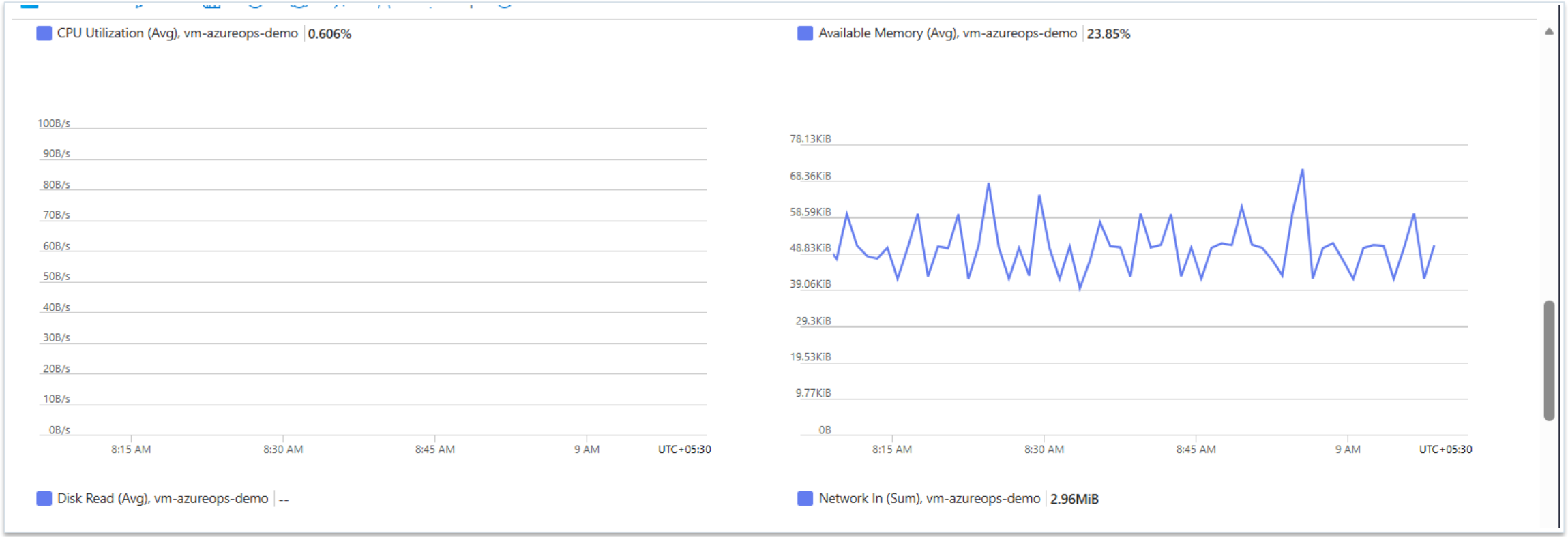
OPERATIONS OVERVIEW





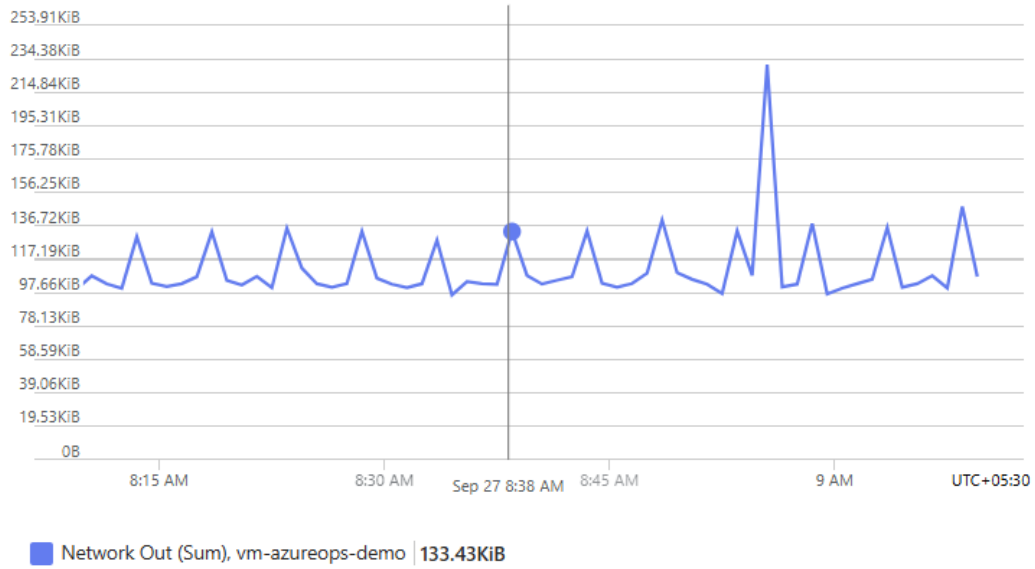
The Azure Operations Dashboard displays monitoring charts and the ACTIVE ALERTS section.

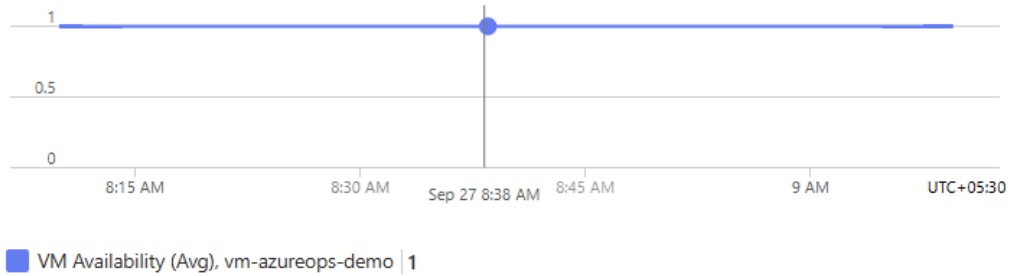
Dashboard — Active Alerts



The dashboard lists Percentage CPU and VM Availability alerts as enabled with severity 3.

Dashboard — CPU, Memory & Network





RESOURCE HEALTH

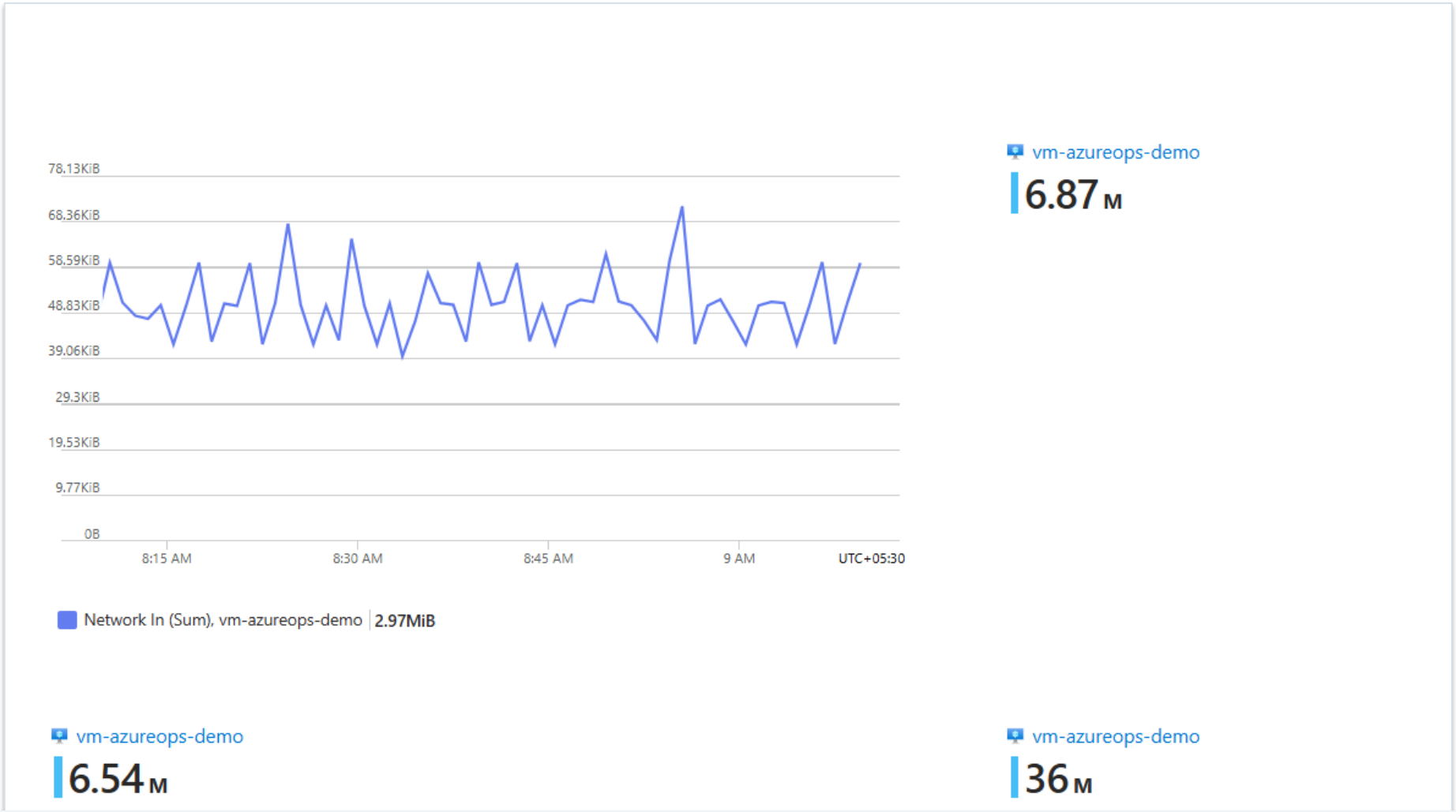
Resource	AvailabilityState	OccurredTime
vm-azureops-demo	Available	2026-09-27T02:00:31.4496666Z

vm-azureops-demo

1

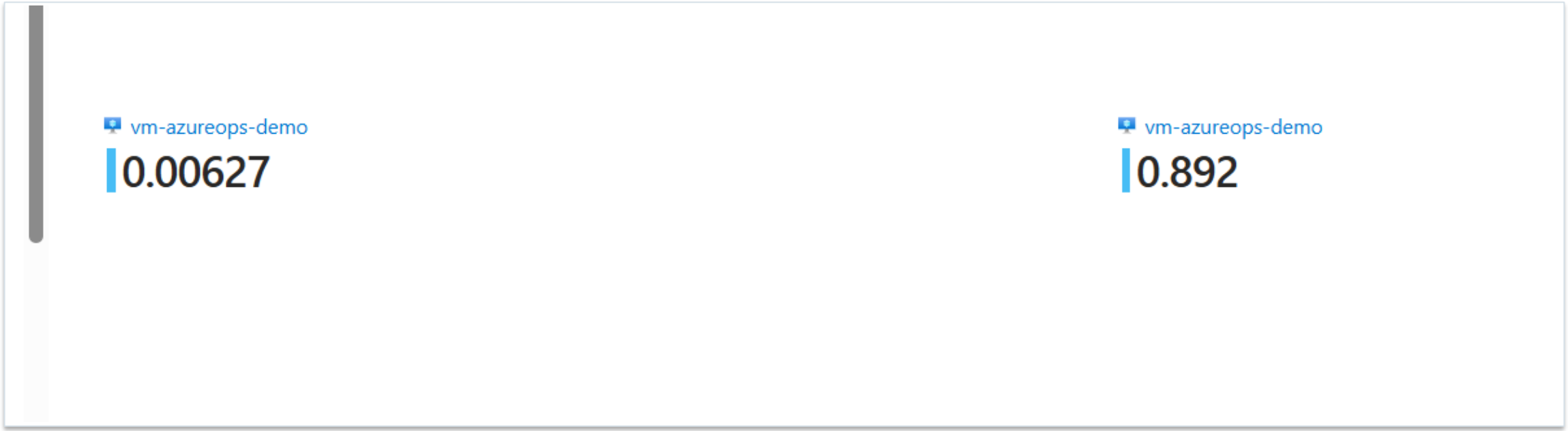
CPU, memory, disk and network charts provide the operational monitoring view.

Dashboard — Resource Health



Resource Health reports vm-azureops-demo as Available, alongside VM availability monitoring.

Dashboard – KPI Values



Compact KPI values provide a quick operational snapshot of network and disk metrics.

Verified Results

VM State	Running
CPU	Minute-level telemetry around 0.6% in captured interval
Memory	Available memory around 23–24%
Network	Inbound and outbound traffic telemetry available
Disk	Write telemetry active; read activity very low
CPU Credits	Continuously reported and increasing
Availability	VM Availability Metric = 1.0
Alerts	2 enabled metric alerts; severity 3 shown
Resource Health	vm-azureops-demo = Available
Workbook	Charts, alerts, KPIs and Resource Health rendered

Conclusion & Scope

- Azure VM health and performance telemetry was successfully collected and visualized.
- The operational workbook provides a consolidated view of CPU, memory, network, disk, alerts and resource health.
- Two metric alerts are enabled for CPU utilization and VM availability.
- The evidence confirms the VM remained available during the captured monitoring period.
- This presentation documents the supplied evidence; it is not a long-term production performance baseline.

Azure Operations Dashboard | rg-azureops-hackathon | vm-azureops-demo